

CONCORD v0.3 Error and Conflict Code Catalog

Portable code catalog with agent-action guidance

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Purpose: This document standardizes error and conflict codes so that different agent implementations do not need to invent their own behavior mapping. Each code carries operational severity and response guidance.

1. Catalog contract

Every error or conflict code in a CONCORD implementation shall carry structured metadata. At minimum, each code entry shall include category, severity, retryable, retry_delay_hint_ms, agent_should, human_likely_needed, blocks_other_actions, and requires_context_refresh.

Metadata field	Allowed values / meaning
severity	informational, warning, elevated, critical
retryable	boolean
retry_delay_hint_ms	Integer or null
agent_should	retry, recheck, queue, wait, escalate, abort, compensate
human_likely_needed	boolean
blocks_other_actions	boolean
requires_context_refresh	boolean

2. Core conflict codes

Code	Severity	Agent should	Typical meaning
STALE_VERSION	warning	recheck	Mutation attempted from an out-of-date authoritative version
SESSION_STALE_VIEW	warning	recheck	Session view is older than its own recorded watermark
LEASE_HELD	warning	wait	Exclusive or scoped lease is held by another session
QUEUE_REQUIRED	informational	queue	Resource requires queue admission before action
QUEUE_AVAILABLE	informational	queue	Queue path exists and should be used
DUPLICATE_INTENT	informational	abort	Equivalent intent already admitted or committed
POLICY_BLOCKED	elevated	escalate	Policy prohibits

			requested action in current context
BUDGET_EXCEEDED	warning	wait	Budget threshold exceeded at gateway or intent layer
CIRCUIT_OPEN	elevated	wait	Circuit breaker is open for this action path
QUORUM_INCOMPLETE	informational	wait	Required quorum has not been met
DEPENDENCY_PENDING	informational	wait	Predecessor dependency has not completed
CAPACITY_EXHAUSTED	warning	wait	No capacity tokens remain for the action path
NEGOTIATION_AVAILABLE	informational	escalate	Negotiation is permitted but not required
NEGOTIATION_REQUIRED	elevated	escalate	Explicit negotiation or arbitration must occur
ESCALATION_REQUIRED	elevated	escalate	Human or privileged process must intervene
AUTHORITATIVE_RECHECK_REQUIRED	warning	recheck	Authoritative source must be re-read before mutation

3. Core error codes

Code	Severity	Agent should	Typical meaning
MISSING_REQUIRED_FIELD	warning	abort	Required contract field absent
INVALID_FIELD_VALUE	warning	abort	Field present but invalid
READ_FRESHNESS_UNAVAILABLE	warning	wait	Required fresh read could not be guaranteed
STALE_READ_WARNING	informational	recheck	Projection lag exceeds declared tolerance
RESOURCE_LOCKED	warning	wait	General lock or resource hold blocks action
NOT_AUTHORIZED_FOR_AGENT_CLASS	elevated	abort	AgentClass lacks capability or trust tier
OVERRIDE_DENIED	elevated	abort	Restrictive override denies the action
SAGA_TIMED_OUT	elevated	compensate	Saga exceeded allowed time policy
SAGA_POISONED	critical	escalate	Saga is no longer automatically recoverable
COMPENSATION_FAILED	critical	escalate	Compensation attempt failed

4. Code metadata examples

Code	retryable	retry_delay_hint_ms	human_likely_needed	blocks_other_actions	requires_context_refresh
STALE_VERSION	True	0	False	False	True
LEASE_HELD	True	1500	False	True	True
BUDGET_EXCEEDED	True	As configured	False	False	False
POLICY_BLOCKED	False	null	True	True	False
SAGA_POISONED	False	null	True	True	True

5. Response construction rules

- Responses shall return one primary code and may include secondary codes where multiple conditions are relevant.
- Where retryable is true, retry_delay_hint_ms should be populated when a meaningful delay can be recommended.
- When requires_context_refresh is true, agents should refresh OperationContext before attempting a new plan.
- When human_likely_needed is true, user-facing systems should display an escalation path rather than silently retry.

6. Minimal JSON response shape

```
{"code": "STALE_VERSION", "category": "conflict", "severity": "warning", "retryable": true, "retry_delay_hint_ms": 0, "agent_should": "recheck", "human_likely_needed": false, "blocks_other_actions": false, "requires_context_refresh": true, "message": "Mutation attempted from an out-of-date version."}
```